

VOLUME 11

COMPLETE VOLUME — CHAPTERS 11.1–11.8

Project Management

Roadmap, Milestones, Sprints, Risk, Changelog, and Trackers for Implementation

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Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Governed By	Volume 0 — Project Foundation; grounded in Volumes 1, 3, 4, 5, 8, 9, 10

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This volume operationalizes the project owner's 13-step implementation sequence: a roadmap mapping each step to the volumes it draws on, binary milestone gates, a lightweight sprint cadence built on Volume 1's own backlog, a consolidated risk register, and the changelog/feature/bug tracking discipline that carries the project through and past release.

Chapter	Title
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11.2	Milestones
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11.4	Risk Register
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CHAPTER 11.1

Development Roadmap

The 13-Step Implementation Sequence, Mapped Against Every Volume It Draws On

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Governed By	Volume 0, Chapter 0.2 — no code before every foundation volume is approved.

1. The Implementation Sequence

This is the 13-step build order the project owner has already set for after Volume 12 is approved. Each step is listed here with the specific volumes/chapters it implements against, so 'start implementation' means executing this table, not re-deriving what to build from scratch.

#	Step	Implements
1	Initialize Project	Volume 7, Chapter 7.11 (Project Initialization) — the literal scaffold from Volume 3, Chapter 3.6/Volume 6, Chapter 6.2.
2	Configure Environment	Volume 7 in full — machine setup through environment variables.
3	Configure Backend	Volume 4 in full — Aurora, Lambda, API Gateway, Firebase provisioning per Volume 7, Chapters 7.7/7.8.
4	Build Authentication	Volume 4, Chapters 4.7/4.8; Volume 6, Chapters 6.4/6.7; Volume 8, Chapter 8.5.
5	Build Core Architecture	Volume 3 and Volume 6 in full — layers, modules, routing, DI, local database, before any feature-specific screen.
6	Build Recording Engine	Volume 5, Chapters 5.1–5.8.
7	Build Upload Engine	Volume 5, Chapters 5.9–5.15.
8	Build UI	Volume 2 (screens/wireframes/design system) implemented as real Flutter widgets per Volume 6's architecture.
9	Integrate APIs	Wiring Volume 3/6's repositories to the real Volume 4 backend, replacing the fakes used in Volume 9, Chapter 9.6/9.7's tests.
10	Test Everything	Volume 9 in full — unit, integration, manual, device matrix.
11	Optimize Performance	Volume 9, Chapter 9.4's targets, measured and tuned against the real build.
12	Internal Testing	Volume 9, Chapter 9.8's UAT — the project owner's sign-off gate.
13	Production Release	Volume 10 in full — the release runbook, Chapter 10.7.

2. Phase Alignment (Volume 1, Chapter 1.11)

Steps 1–13 above build the MVP scope exactly as Volume 1 fixed it (Admin/Collector core, comprehensive metadata) — Phase 2 (secondary Collector features), Phase 3 (advanced Admin/Ops tooling), and Phase 4 (future capabilities) are not part of this initial 13-step sequence and begin only after Production Release, each following the same volume-then-build discipline for whatever new requirements they introduce.

3. Sequencing Dependencies

Why this exact order: each step depends on the one before it structurally, not arbitrarily — Authentication (4) can't be built without Core Architecture's DI graph (5) to hold it, but is listed before Core Architecture here only as a roadmap-readability choice; Chapter 11.3's sprint plan is where the real, revisable execution order (which may interleave 4 and 5) is defined in practice.

- Recording Engine (6) and Upload Engine (7) both depend on Core Architecture (5) and Authentication (4), but not on each other's completion — they can be built in parallel by two contributors if the team ever grows beyond one.
- UI (8) depends on Volume 2's screens existing, which they already do; it does not depend on Recording/Upload Engine completion for its non-recording screens (Projects/Tasks/Admin), only for C-09/C-11/C-12 specifically.

4. What This Chapter Defers

- Concrete milestone exit criteria for each step — Chapter 11.2.
- Sprint-level breakdown and cadence — Chapter 11.3.

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CHAPTER 11.2

Milestones

Exit Criteria for Each of the 13 Implementation Steps

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1. Milestone Table

Each milestone below is a binary, checkable gate — the project owner can look at this table at any point during implementation and know exactly where the build stands, without needing to ask.

Milestone	Exit Criteria
M1 — Environment Ready	Volume 7, Chapter 7.13's verification checklist passes in full on the primary Windows machine.
M2 — Backend Live	Every Volume 4 endpoint (Chapter 4.6) deployed and reachable in the dev environment; Aurora schema (Chapter 4.4) migrated.
M3 — Auth Working	A test Admin and test Collector account can both log in (SH-02) and reach their correct root (Role Router, Chapter 2.4 §4) against the real backend.
M4 — Core Architecture Scaffolded	Volume 3/6's layers, modules, routing, and DI graph exist and compile with placeholder screens; Volume 6, Chapter 6.5's Drift schema is live locally.
M5 — Recording Engine Functional	A real device (Volume 7, Chapter 7.1) records a full session, produces correctly-chunked, checksummed local files per Volume 5, Chapters 5.1–5.8.
M6 — Upload Engine Working E2E	A recorded chunk uploads to the real dev S3 bucket, verifies via the real backend, and reaches Complete — Volume 5, Chapters 5.9–5.15 fully wired to Volume 4.
M7 — UI Complete	Every screen in Volume 2, Chapter 2.5's inventory is built and matches the Design System (Chapter 2.8) and Wireframes (Chapter 2.6).
M8 — APIs Integrated	Every repository (Volume 3, Chapter 3.4) reads/writes the real backend — no fake/mock repository remains wired into a release build.
M9 — Test Suite Green	Volume 9, Chapters 9.6/9.7's full suite passes in CI (Volume 7, Chapter 7.13) at the coverage targets Chapter 9.5 sets.
M10 — Performance Targets Met	Every target in Volume 9, Chapter 9.4 measured and met on the reference device.
M11 — Internal Testing Passed	Volume 9, Chapter 9.8's manual test plan and UAT sign-off complete across Chapter 9.9's device matrix.
M12 — Store-Ready	Volume 10, Chapters 10.1–10.4's build, signing, and store-listing requirements all satisfied for both platforms.
M13 — Production Release	Volume 10, Chapter 10.7's runbook executed through 100% rollout with no Chapter 10.6 rollback triggered.

2. Using This Table

A milestone is not marked complete until its exit criteria is fully met — partial completion (e.g. 'most screens built' for M7) is tracked as in-progress against Chapter 11.3's sprint backlog, not prematurely checked off here. This table is the single source of truth for 'how far along are we,' referenced rather than re-derived in status updates.

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CHAPTER 11.3

Sprint Planning

A Lightweight Cadence Sized for This Team

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Chapter Reference	11.3
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1. Cadence

Two-week sprints, each aligned to one or more of Chapter 11.1's 13 steps rather than an arbitrary story-point target — since the steps themselves are already ordered and dependency-checked, a sprint's job is simply to decide how much of the current step(s) fit in two weeks, not to re-plan sequencing from scratch each time.

2. Backlog Source

The backlog is Volume 1's user stories, not a separately invented one: US-01 through US-33 (Volume 1, Chapter 1.7) are the actual backlog items — each already carries its FR/BR traceability, so a sprint's stories are pulled directly from there rather than re-written. Chapter 11.6's Feature Tracker is where each story's implementation status lives.

3. Ceremonies (Lightweight)

Ceremony	Frequency	Purpose
Sprint planning	Start of each sprint	Select which US-* stories / roadmap step(s) (Ch.11.1) this sprint targets.
Progress check-in	Weekly (mid-sprint)	Informal — flag blockers early rather than waiting for sprint end.
Sprint review	End of each sprint	Check completed work against Chapter 11.2's milestone exit criteria — did anything reach a milestone?
Retro	End of each sprint, brief	What slowed this sprint down; feed into the Risk Register (Ch.11.4) if it's a recurring pattern.

4. Definition of Ready

- A story is ready to pull into a sprint only once its FR/BR/US traceability is clear (already true for every Volume 1 story) and any architecture it depends on (Volume 3/6) is already built — a story is never started ahead of the architectural foundation it needs, mirroring Volume 0's own layered-approval philosophy at the sprint level.

5. Definition of Done

Deliberately not re-defined here — Chapter 12.10 (Volume 12) is the single authoritative Definition of Done every story is held to, so this chapter doesn't risk drifting out of sync with a second, competing definition.

6. Solo/Small-Team Adjustment

With Faisal as the primary owner (and Claude as an implementation collaborator, per Volume 12), ceremonies are intentionally minimal — no separate scrum master role, no story-pointing ritual

beyond a rough t-shirt-size estimate if useful. The cadence exists to create a regular checkpoint against Chapter 11.2's milestones, not to impose process for its own sake.

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CHAPTER 11.4

Risk Register

Every Risk Flagged Since Volume 3, Consolidated in One Place

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Chapter Reference	11.4
Version	1.0
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Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
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1. Purpose

Individual volumes flagged specific risks as they arose (a lens-selection fallback here, an iOS background-execution limit there). This chapter is the first place they're all visible together, so nothing quietly gets forgotten between now and implementation.

2. Technical Risks

Risk	Likelihood	Impact	Mitigation
Wide-angle lens fragmentation across Android devices	High	High	Volume 5, Chapter 5.1's 3-tier capability ladder; Volume 9, Chapter 9.9's manufacturer-variant device matrix catches gaps before release.
iOS background upload time limits	Medium	Medium	Volume 5, Chapter 5.11 — falls back to Queued/Failed safely, never loses data (BR-08); Volume 0's iOS-secondary priority accepts this as a known, bounded limitation.
Lambda cold-start latency	Medium	Low	Accepted tradeoff per ADR-007 (Volume 4); provisioned concurrency is the named mitigation if measured latency requires it (Volume 3, Chapter 3.1).
Aurora/S3 (AWS) + Firebase (Google) multi-cloud seam	Low	Medium	Deliberately narrow — only token verification crosses the boundary (Volume 4, Chapter 4.9, Section 3).
Generated-code drift (contributor forgets build_runner)	Medium	Low	CI drift check (Volume 7, Chapter 7.13) fails the build automatically.

3. Product/Compliance Risks

Risk	Likelihood	Impact	Mitigation
Bystander footage captured incidentally	High (structural — inherent to egocentric field capture)	High	Flagged explicitly in Volume 8, Chapter 8.6 as an operational/legal question for Human Archive, not solvable by engineering alone — recommends legal review and field-notice practices.
Privacy/retention policy not matching actual regulatory requirements	Medium	High	Volume 8, Chapters 8.6/8.7 both explicitly flagged for legal review before publication — not treated as final without it.
App Store rejection over location-permission justification	Medium	Medium	Volume 10, Chapter 10.4's specific, narrow usage-string language and reviewer notes address Apple's most common rejection pattern here directly.

4. Project/Delivery Risks

Risk	Likelihood	Impact	Mitigation
Single-developer bus factor (Windows-first, largely solo build)	Medium	High	This entire documentation set (Volumes 0–12) exists specifically so a second person or an AI collaborator (Volume 12) can pick up work with full context — the primary mitigation is the documentation itself.
Scope creep beyond MVP (Volume 1, Chapter 1.10)	Medium	Medium	Chapter 11.1's roadmap and Chapter 11.6's feature tracker only track MVP-scope items until Chapter 11.2's M13 is reached; Phase 2/3/4 items are explicitly out of the current sprint backlog (Ch.11.3).

Risk	Likelihood	Impact	Mitigation
Documentation drift from actual code once implementation starts	Medium	Medium	Volume 12, Chapter 12.4's documentation rules make this an explicit, enforced discipline for any AI-assisted implementation work.

5. Review Cadence

This register is reviewed at every sprint retro (Chapter 11.3) — a new risk surfacing during implementation is added here, not left undocumented in someone's memory.

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CHAPTER 11.5

Changelog

A Keep-a-Changelog Template Tied to Volume 10's Versioning Scheme

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Chapter Reference	11.5
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1. Format

Follows the Keep a Changelog convention, grouped under an Unreleased section until a version is actually shipped, then dated. Versions are the same MAJOR.MINOR.PATCH versionName fixed in Volume 10, Chapter 10.1 — this chapter does not introduce a second numbering scheme.

2. Entry Categories

Category	Meaning
Added	A new capability — typically a Volume 1 FR/US shipping for the first time.
Changed	A behavior change to something already shipped.
Fixed	A bug fix — cross-reference the Bug Tracker (Chapter 11.7) entry ID.
Security	Anything touching auth, storage, or data handling (Volume 8).
Deprecated / Removed	A capability being phased out or removed — rare before v1.0.0.

3. Template

```
## [Unreleased]
### Added
- ...

## [1.0.0] - YYYY-MM-DD
### Added
- Initial production release (Volume 10, Chapter 10.7).
```

4. Rules

- Every entry is written the day the change merges, not batched at release time — batching is the single most common way changelogs go stale (Chapter 11.8's maintenance discipline applies here directly).
- An entry links back to its FR/US/Bug ID wherever one exists, so the changelog and the Feature/Bug Trackers (Chapters 11.6/11.7) never tell two different stories about the same change.
- Pre-1.0.0 development builds may log only against '[Unreleased]' — the first dated entry is v1.0.0 itself, at Milestone M13 (Chapter 11.2).

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CHAPTER 11.6

Feature Tracker

Every FR Group from Volume 1, Mapped to Build Status

Document Title	Feature Tracker
Chapter Reference	11.6
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026

1. Purpose

Volume 1, Chapter 1.3 fixed every functional requirement as FR-AUTH, FR-ONB, FR-ADM, FR-PT, FR-CHK, FR-REC, FR-CHNK, FR-META, FR-UPL, FR-SES, and FR-SEC. This tracker is where each group's actual build status lives — it is not re-derived elsewhere, and Chapter 11.3's sprint backlog updates it, not the reverse.

2. Status Legend

Not Started / In Progress / Built (unverified) / Verified (Volume 9 tests passing) / Shipped (in a released version, Chapter 11.5).

3. Tracker

FR Group (Volume 1, Ch.1.3)	Status	Notes
FR-AUTH — Authentication & Session	Not Started	Depends on Roadmap Step 4 (Ch.11.1); Volume 4, Ch.4.7/4.8 + Volume 8, Ch.8.5.
FR-ONB — Onboarding & Permissions	Not Started	Depends on Core Architecture (Step 5) and Volume 6, Ch.6.4.
FR-ADM — Admin Project & Task Mgmt	Not Started	UI (Step 8) against Volume 2's Admin screens; repositories per Volume 3, Ch.3.4.
FR-PT — Collector Dashboard/Projects/Tasks	Not Started	UI (Step 8); offline cache per Volume 6, Ch.6.5 (Drift).
FR-CHK — Pre-Recording Checklist	Not Started	Volume 5, Ch.5.1's capability ladder gates this.
FR-REC — Recording	Not Started	Roadmap Step 6; Volume 5, Ch.5.1–5.8 — the core-promise feature (Volume 0).
FR-CHNK — Chunking & Local Processing	Not Started	Volume 5, Ch.5.3's internal-Stop mechanism; Step 6.
FR-META — Metadata Generation & Integrity	Not Started	Volume 5, Ch.5.7/5.8; checksum discipline verified in Volume 9, Ch.9.6.
FR-UPL — Upload	Not Started	Roadmap Step 7; Volume 5, Ch.5.9–5.15 wired to Volume 4 backend.
FR-SES — Session Lifecycle & Status	Not Started	Cuts across Recording/Upload; status surfaces on Collector Dashboard (FR-PT).
FR-SEC — Secondary Features	Not Started	Explicitly Phase 2+ per Volume 1, Ch.1.10/1.11 — out of the initial 13-step sequence (Ch.11.1 §2).

4. Update Discipline

A row moves to 'Verified' only once its Volume 9 test coverage (Chapters 9.6/9.7) actually passes in CI — not on the developer's local machine. A row moves to 'Shipped' only once Chapter 11.5 has a dated changelog entry for it. No row skips a status.

CHAPTER 11.7

Bug Tracker

Template, Process, and Severity Levels Aligned to Volume 10's Crash Triage

Document Title	Bug Tracker
Chapter Reference	11.7
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1. Severity Levels

Reuses the same risk-weighting logic Volume 10, Chapter 10.6 applies to crash triage — a bug in Recording or Upload (Volume 5) outranks an equally-frequent bug in a secondary screen, because Volume 0 fixes those two flows as the core promise.

Severity	Definition	Response Target
S1 — Critical	Data loss, or Recording/Upload (Volume 5) broken or silently failing.	Same business day.
S2 — High	A Must-Have FR (Volume 1, Ch.1.3) broken outside Recording/Upload, no workaround.	Within 2 business days.
S3 — Medium	A Should-Have FR broken, or a Must-Have with a workaround.	Next sprint (Ch.11.3).
S4 — Low	Cosmetic, or an edge case with negligible user impact.	Backlog, unscheduled.

2. Bug Report Template

```
ID: BUG-XXX
Title:
Severity: S1 / S2 / S3 / S4
Affected FR/US (Volume 1):
Steps to Reproduce:
Expected vs Actual:
Device/Environment (Volume 9, Ch.9.9 matrix):
Linked Crashlytics event (if any, Volume 10 Ch.10.6):
```

3. Process

- Every bug is logged here before it's fixed, even a five-minute fix — the tracker is the record of what broke, not just what's still open.
- An S1 bug blocks the current sprint's other work until resolved or explicitly deferred with the project owner's sign-off (mirrors Volume 0's approval discipline at the bug level).
- A fix's changelog entry (Chapter 11.5, 'Fixed') references the BUG-XXX ID — the two records stay linked, never duplicated with different wording.

4. Closing Criteria

A bug closes only once its regression is covered by a Volume 9 automated test where feasible (Chapter 9.6/9.7) — closing on manual verification alone is acceptable only for S3/S4 UI-only issues.

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CHAPTER 11.8

Maintenance Guide

What Happens After Production Release (M13)

Document Title	Maintenance Guide
Chapter Reference	11.8
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1. Purpose

Volume 10 ends at Production Release. This chapter is what keeps the app healthy afterward — it closes the loop the roadmap (Chapter 11.1) opens, so the documentation set doesn't end the moment shipping does.

2. Recurring Cadences

Activity	Cadence	Reference
Dependency updates (Flutter/Dart, npm packages)	Monthly, or immediately for a security advisory	Volume 3, Chapter 3.8's fvm-pinned upgrade policy.
Monitoring dashboard review	Weekly	Volume 10, Chapter 10.5's metrics — checked against Volume 0's own success criteria, not just 'no red alerts.'
Crash triage	Daily on new clusters	Volume 10, Chapter 10.6.
Risk Register review	Every sprint retro	Chapter 11.4.
Documentation accuracy check	Every release	Section 3 below.
Store listing / compliance re-check	Each OS major version bump	Volume 10, Chapters 10.3/10.4.

3. Documentation-Drift Discipline

- A code change that alters behavior described in Volumes 1–10 updates the affected chapter in the same pull request, not as a follow-up 'documentation task' — Volume 12, Chapter 12.4 makes this an enforced rule for AI-assisted changes specifically.
- The Feature Tracker (Chapter 11.6) and Changelog (Chapter 11.5) are updated as part of the definition of done (Volume 12, Chapter 12.10) for every merged change, not on a separate cadence.

4. Ownership

Faisal (Human Archive) remains the sole approval authority for scope changes post-launch, consistent with Volume 0's Constitution — this guide governs routine upkeep, not a change to that authority.

5. What This Chapter Does Not Cover

Phase 2/3/4 feature development (Volume 1, Chapter 1.11) is new scope, not maintenance — it re-enters the Volume 0 discipline (a proposal, requirements, approval) rather than being treated as upkeep under this chapter.